

Final-Series02Rstudio

Zoé, Elouan, Tuba

01 October 2026, 22:28:09

2 Statistical tests: comparison of medians (or means?)

2.1 Make clear what is a statistical test, and when to use it.

What is a median? A mean? The mean is the sum of all values divided by their number, and the median is the middle value once the data are sorted. *Reference: Field, A. (2013). Discovering Statistics Using IBM SPSS Statistics, Chapter 2.*

What is a variance? A standard deviation? The variance is the average squared deviation from the mean, and the standard deviation is its square root, expressed in the same unit as the data. *Reference: Field, A. (2013). Discovering Statistics Using IBM SPSS Statistics, Chapter 2.*

What is a normal distribution? A normal distribution is a symmetric, bell-shaped distribution in which most values cluster around the mean. *Reference: STHDA - Normality Test in R*

What is a statistical test? A statistical test uses sample data to decide whether there is enough evidence to reject a null hypothesis, by computing a p-value. *Reference: STHDA - Statistical Tests and Assumptions*

When to use a statistical test? A statistical test is used to know whether a result observed on a sample is likely real or due to random sampling variation. *Reference: STHDA - Statistical Tests and Assumptions*

What is a parametric test? A non-parametric test? A parametric test assumes a specific distribution of the data (usually normal), whereas a non-parametric test makes no such assumption and typically works on ranks. *Reference: STHDA - Non-parametric Tests*

What are the assumptions of parametric tests? Parametric tests generally assume normally distributed data, independent observations and, for group comparisons, equal variances. *Reference: Field, A. (2013). Discovering Statistics Using IBM SPSS Statistics, Chapter 5.*

What are the assumptions of non-parametric tests? Non-parametric tests mainly assume independent observations and data that can be ranked. *Reference: STHDA - Non-parametric Tests*

What is a p-value? The p-value is the probability of obtaining a result at least as extreme as the one observed if the null hypothesis is true. *Reference: STHDA - Statistical Tests and Assumptions*

What is the risk of error when using a statistical test? A Type I error is a false positive (risk usually set at $\alpha = 0.05$), and a Type II error is a false negative, often due to low statistical power. *Reference: Field, A. (2013). Discovering Statistics Using IBM SPSS Statistics, Chapter 2.*

What is the difference between a paired and an unpaired test? A paired test compares two measurements from the same subjects, whereas an unpaired test compares two independent groups. *Reference: STHDA - t-test Formula*

2.2 Effect or treatment over time

2.2.1 The data

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.3
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.2
✓ ggplot2    4.0.2    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts — tidyverse_conflicts() —
```

```
* dplyr::filter() masks stats::filter()
* dplyr::lag() masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
prepost <- read_csv("data/PrePost.csv")
```

```
Rows: 16 Columns: 2
— Column specification —————
Delimiter: ","
chr (1): time
dbl (1): perf

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(prepost)
```

```
Rows: 16
Columns: 2
$ perf <dbl> 125, 130, 132, 135, 136, 138, 140, 145, 127, 132, 133, 136, 139, ...
$ time <chr> "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before", "Before"
```

```
summary(prepost)
```

```
      perf      time
Min.   :125.0  Length:16
1st Qu.:132.0  Class  :character
Median :136.0  Mode   :character
Mean    :136.4
3rd Qu.:140.2
Max.    :148.0
```

```
count(prepost, time)
```

```
# A tibble: 2 × 2
  time      n
  <chr> <int>
1 After     8
2 Before    8
```

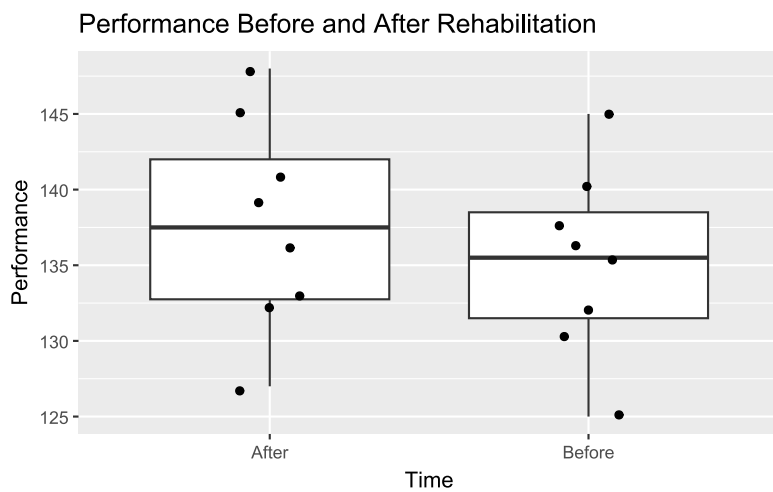
The dataset contains 16 observations: 8 measurements before rehabilitation and 8 measurements after rehabilitation. The variable time identifies the measurement period (Before or After), while perf represents the measured performance variable.

Variable	Description	Unit	Possible values	Expected change
perf	Cardiac rehabilitation performance score; the exact test/measure is not specified in the assignment materials	Not specified (likely arbitrary or composite score)	125–148 observed in this dataset	Expected to improve after rehabilitation
time	Measurement period	Categorical	Before / After	Not applicable

2.2.2 The analysis

```
ggplot(prepost, aes(x = time, y = perf)) +
  geom_boxplot() +
  geom_jitter(width = 0.1) +
```

```
labs(
  title = "Performance Before and After Rehabilitation",
  x = "Time",
  y = "Performance"
)
```



```
prepost
```

```
# A tibble: 16 × 2
  perf time
<dbl> <chr>
1  125 Before
2  130 Before
3  132 Before
4  135 Before
5  136 Before
6  138 Before
7  140 Before
8  145 Before
9  127 After
10 132 After
11 133 After
12 136 After
13 139 After
14 141 After
15 145 After
16 148 After
```

```
before <- prepost$perf[prepost$time == "Before"]
after <- prepost$perf[prepost$time == "After"]
```

```
difference <- after - before
```

```
difference
```

```
[1] 2 2 1 1 3 3 5 3
```

```
shapiro.test(difference)
```

Shapiro-Wilk normality test

```
data: difference
W = 0.89896, p-value = 0.2828
```

A paired t-test was used because the same participants were measured before and after rehabilitation. The distribution of the paired differences did not show evidence of departure from normality according to the Shapiro-Wilk test ($p = 0.283$).

```
t.test(after, before, paired = TRUE)
```

Paired t-test

```
data: after and before
t = 5.4006, df = 7, p-value = 0.001008
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.405392 3.594608
sample estimates:
mean difference
          2.5
```

The paired t-test showed a statistically significant increase in performance after rehabilitation ($t(7) = 5.40$, $p = 0.001$). Performance increased by an average of 2.5 units, with a 95% confidence interval from 1.41 to 3.59 units. These results suggest that performance improved after the rehabilitation program.

2.3: Testing some stereotypes

Loading the data

```
# Import the dataset
Snore <- read.delim("snore.txt", header = TRUE)

head(Snore)
```

	age	weight	height	alcohol	sex	snore	tabac
1	47	71	158	0	H	N	0
2	56	58	164	7	H	0	N
3	46	116	208	3	H	N	0
4	70	96	186	3	H	N	0
5	51	91	195	2	H	0	0
6	46	98	188	0	F	N	N

```
str(Snore) # structure of the dataset
```

```
'data.frame': 100 obs. of 7 variables:
 $ age : int 47 56 46 70 51 46 40 46 49 39 ...
 $ weight: int 71 58 116 96 91 98 112 77 76 119 ...
 $ height: int 158 164 208 186 195 188 193 165 164 196 ...
 $ alcohol: int 0 7 3 3 2 0 5 0 0 3 ...
 $ sex : chr "H" "H" "H" "H" ...
 $ snore : chr "N" "0" "N" "N" ...
 $ tabac : chr "0" "N" "0" "0" ...
```

```
summary(Snore)
```

	age	weight	height	alcohol
Min.	:23.00	Min. : 42.00	Min. :158.0	Min. : 0.00
1st Qu.:	43.00	1st Qu.: 77.00	1st Qu.:166.0	1st Qu.: 0.00
Median :	52.00	Median : 95.00	Median :186.0	Median : 2.00
Mean :	52.27	Mean : 90.41	Mean :181.1	Mean : 2.95

```

3rd Qu.:62.25   3rd Qu.:107.00   3rd Qu.:194.0   3rd Qu.: 4.25
Max.      :74.00   Max.      :120.00   Max.      :208.0   Max.      :15.00
      sex              snore              tabac
Length:100      Length:100      Length:100
Class :character Class :character Class :character
Mode  :character Mode  :character Mode  :character

```

```

# Check the categories of categorical variables
unique(Snore$sex)

```

```
[1] "H" "F"
```

```
unique(Snore$snore)
```

```
[1] "N" "O"
```

```
unique(Snore$tabac)
```

```
[1] "O" "N"
```

```

# Count observations in each category
table(Snore$sex)

```

```

 F  H
25 75

```

```
table(Snore$snore)
```

```

 N  O
65 35

```

```
table(Snore$tabac)
```

```

 N  O
36 64

```

First stereotype: Are snorers fatter?

```
### Are snorers fatter?
```

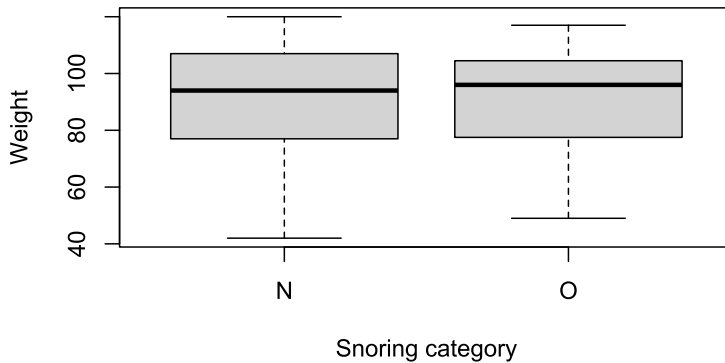
```
# Compare weight between snorers and non-snorers
```

```

boxplot(weight ~ snore,
        data = Snore,
        xlab = "Snoring category",
        ylab = "Weight",
        main = "Weight comparison between snorers and non-snorers")

```

Weight comparison between snorers and non-snorers



```
# Calculate median weight in each group
tapply(Snore$weight, Snore$snore, median)
```

```
 N  O
94 96
```

```
# Statistical analysis
# Check normality of weight in each snoring group
shapiro.test(Snore$weight[Snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data:  Snore$weight[Snore$snore == "N"]
W = 0.95527, p-value = 0.01955
```

```
shapiro.test(Snore$weight[Snore$snore == "O"])
```

Shapiro-Wilk normality test

```
data:  Snore$weight[Snore$snore == "O"]
W = 0.92744, p-value = 0.0235
```

```
# Compare weight between the two independent groups
wilcox.test(weight ~ snore,
             data = Snore,
             exact = FALSE)
```

Wilcoxon rank sum test with continuity correction

```
data:  weight by snore
W = 1154, p-value = 0.9079
alternative hypothesis: true location shift is not equal to 0
```

Result – Weight comparison The median weight was 94 among non-snorers and 96 among snorers. The Wilcoxon rank-sum test showed no statistically significant difference in weight between the two groups ($W = 1154$, $p = 0.9079$). Therefore, this dataset does not provide statistical evidence supporting the stereotype that snorers are heavier.

Second Stereotype: Are snorers drinking or smoking more?

```
# Compare alcohol categories between snorers and non-snorers
table(Snore$snore, Snore$alcohol)
```

```
   0  1  2  3  4  5  6  7  8 10 12 15
N 33  3  3  7  7  3  0  1  4  3  1  0
0  9  0  3  3  7  3  2  2  4  1  0  1
```

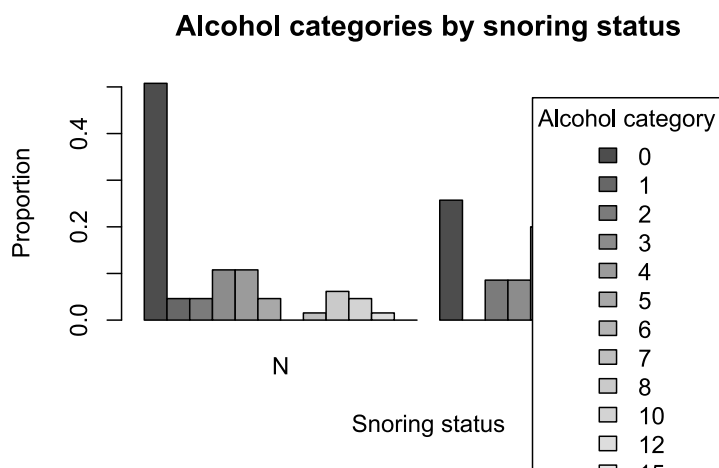
```
# Calculate the proportion of each alcohol category within each snoring group
alcohol_prop <- prop.table(table(Snore$snore, Snore$alcohol), 1)

# Display proportions
alcohol_prop
```

```
   0         1         2         3         4         5
N 0.50769231 0.04615385 0.04615385 0.10769231 0.10769231 0.04615385
0 0.25714286 0.00000000 0.08571429 0.08571429 0.20000000 0.08571429

   6         7         8        10        12        15
N 0.00000000 0.01538462 0.06153846 0.04615385 0.01538462 0.00000000
0 0.05714286 0.05714286 0.11428571 0.02857143 0.00000000 0.02857143
```

```
# Visualise alcohol categories by snoring status
barplot(t(alcohol_prop),
        beside = TRUE,
        legend.text = TRUE,
        args.legend = list(title = "Alcohol category"),
        xlab = "Snoring status",
        ylab = "Proportion",
        main = "Alcohol categories by snoring status")
```



```
# Calculate mean and median alcohol values for each group
tapply(Snore$alcohol, Snore$snore, mean)
```

```
   N         0
2.369231 4.028571
```

```
tapply(Snore$alcohol, Snore$snore, median)
```

```
N 0
0 4
```

```
# Statistical analysis
# Check normality of alcohol values in each snoring group
shapiro.test(Snore$alcohol[Snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data: Snore$alcohol[Snore$snore == "N"]
W = 0.76256, p-value = 6.818e-09
```

```
shapiro.test(Snore$alcohol[Snore$snore == "0"])
```

Shapiro-Wilk normality test

```
data: Snore$alcohol[Snore$snore == "0"]
W = 0.90387, p-value = 0.005032
```

```
# Compare alcohol values between the two independent groups
wilcox.test(alcohol ~ snore,
            data = Snore,
            exact = FALSE)
```

Wilcoxon rank sum test with continuity correction

```
data: alcohol by snore
W = 788, p-value = 0.008594
alternative hypothesis: true location shift is not equal to 0
```

Result – Alcohol consumption

The mean recorded alcohol value was 2.37 among non-snorers and 4.03 among snorers, while the respective medians were 0 and 4. The Wilcoxon rank-sum test revealed a statistically significant difference between the two groups ($W = 788$, $p = 0.008594$). Snorers therefore had higher recorded alcohol values in this sample.

```
# Are snorers more likely to smoke?

# Compare smoking categories by snoring status
table(Snore$snore, Snore$tabac)
```

```
      N 0
N 21 44
0 15 20
```

```
# Proportion of smoking categories in each snoring group
smoking_prop <- prop.table(
  table(Snore$snore, Snore$tabac), 1
)

# Display proportions
smoking_prop
```


	N	O
N	0.3230769	0.6769231
O	0.4285714	0.5714286

```
# Create a bar plot
barplot(t(smoking_prop),
  beside = TRUE,
  legend.text = c("Non-smokers (N)", "Smokers (O)"),
  args.legend = list(title = "Tobacco status"),
  xlab = "Snoring status",
  ylab = "Proportion",
  main = "Smoking status by snoring status")
```



```
# Statistical analysis
fisher.test(table(Snore$snore, Snore$tabac))
```

Fisher's Exact Test for Count Data

```
data: table(Snore$snore, Snore$tabac)
p-value = 0.3827
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.2516104 1.6282374
sample estimates:
odds ratio
 0.6393445
```

Result – Smoking habits among snorers and non-snorers

Among non-snorers, 67.7% were smokers, compared with 57.1% among snorers. Fisher's exact test is appropriate because both variables are categorical. The test did not reveal a statistically significant association between snoring and smoking status ($p = 0.3827$). Therefore, the data do not provide sufficient statistical evidence to conclude that snorers are more likely to smoke.

Third Stereotype: Are men fatter?

```
# count observations in each sex category
table(Snore$sex)
```

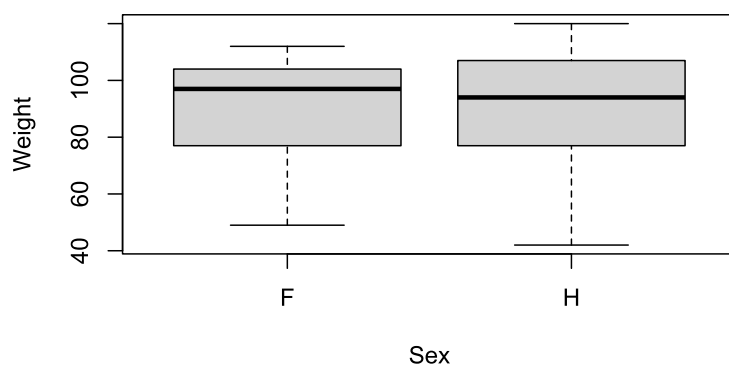
F	H
25	75

```
unique(Snore$sex)
```

```
[1] "H" "F"
```

```
# create the boxplot to look at the difference of both gender's weight
boxplot(weight ~ sex,
        data = Snore,
        xlab = "Sex",
        ylab = "Weight",
        main = "Weight comparison between men and women")
```

Weight comparison between men and women



```
# calculate median weight in each sex category
tapply(Snore$weight, Snore$sex, median)
```

```
F H
97 94
```

```
# Statistical analysis
```

```
# Check normality of weight in women and men
shapiro.test(Snore$weight[Snore$sex == "F"])
```

Shapiro-Wilk normality test

```
data:  Snore$weight[Snore$sex == "F"]
W = 0.9146, p-value = 0.03865
```

```
shapiro.test(Snore$weight[Snore$sex == "H"])
```

Shapiro-Wilk normality test

```
data:  Snore$weight[Snore$sex == "H"]
W = 0.95406, p-value = 0.008379
```

```
# Compare weight between women and men
wilcox.test(weight ~ sex,
            data = Snore,
            exact = FALSE)
```

```
Wilcoxon rank sum test with continuity correction
```

```
data: weight by sex  
W = 893.5, p-value = 0.729  
alternative hypothesis: true location shift is not equal to 0
```

Result – Weight comparison between men and women The median weight was 97 among women and 94 among men. The Wilcoxon rank-sum test showed no statistically significant difference in weight between women and men ($W = 893.5$, $p = 0.729$). Thus, the data do not provide statistical evidence supporting the stereotype that men are heavier than women.

Fourth Stereotype: Do women smoke less?

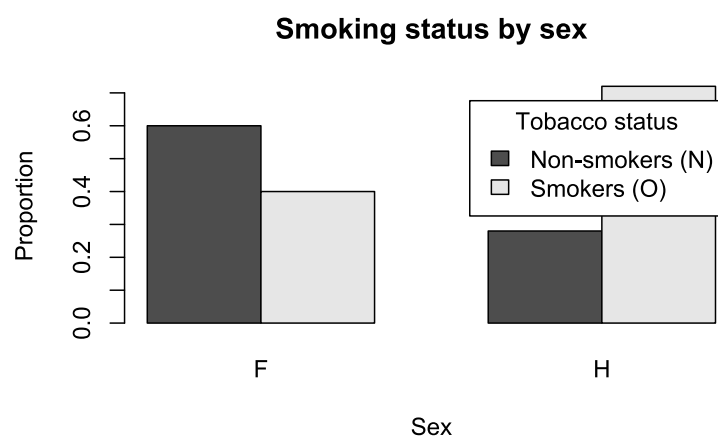
```
# compare smoking status between men and women  
table(Snore$sex, Snore$tabac)
```

```
  N  0  
F 15 10  
H 21 54
```

```
# calculate proportions  
smoking_by_sex <- prop.table(table(Snore$sex, Snore$tabac), 1)  
  
smoking_by_sex
```

```
  N    0  
F 0.60 0.40  
H 0.28 0.72
```

```
# create the bar plot to look at the difference of both gender's smoking status  
barplot(t(smoking_by_sex),  
        beside = TRUE,  
        legend.text = c("Non-smokers (N)", "Smokers (O)"),  
        args.legend = list(title = "Tobacco status"),  
        xlab = "Sex",  
        ylab = "Proportion",  
        main = "Smoking status by sex")
```



```
# Statistical analysis  
# Test the association between sex and smoking status  
fisher.test(table(Snore$sex, Snore$tabac))
```

Fisher's Exact Test for Count Data

```
data: table(Snore$sex, Snore$tabac)
p-value = 0.007276
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.354726 11.139908
sample estimates:
odds ratio
 3.79892
```

Result – Smoking habits among women and men

The proportion of smokers was 40% among women and 72% among men. Fisher's exact test revealed a statistically significant association between sex and smoking status ($p = 0.007276$). These results indicate that women had a lower smoking prevalence than men in this sample, providing statistical support for the stereotype that women smoke less.

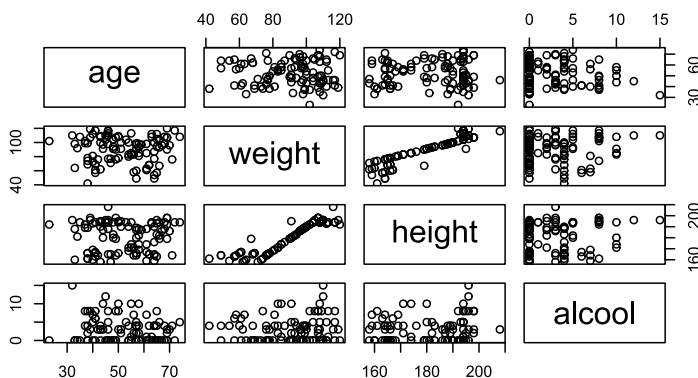
General analysis: Correlations between variables

```
# select numerical variables
numerical_data <- Snore[, c("age", "weight", "height", "alcohol")]

# Calculate correlation matrix for selected numerical variables
cor_matrix <- cor(Snore[, c("age", "weight", "height", "alcohol")],
                  use = "complete.obs")

# visualise pairwise relationships between numerical variables
pairs(Snore[, c("age", "weight", "height", "alcohol")],
      main = "Relationships between numerical variables")
```

Relationships between numerical variables



```
#Statistical analysis: Spearman correlations
# Calculate the Spearman correlation matrix
spearman_matrix <- cor(numerical_data,
                       method = "spearman",
                       use = "complete.obs")

# Display Spearman correlation coefficients
spearman_matrix
```

	age	weight	height	alcohol
age	1.000000000	-0.01455125	0.008072461	-0.10748796
weight	-0.014551254	1.00000000	0.930383489	0.08568687
height	0.008072461	0.93038349	1.00000000	0.12539769
alcohol	-0.107487956	0.08568687	0.125397694	1.00000000

```
# Test each pairwise Spearman correlation
correlation_pairs <- combn(names(numerical_data),
                           2,
                           simplify = FALSE)

for (pair in correlation_pairs) {
  cat("\nCorrelation between", pair[1], "and", pair[2], "\n")

  print(cor.test(numerical_data[[pair[1]]],
                 numerical_data[[pair[2]]],
                 method = "spearman",
                 exact = FALSE))
}
```

Correlation between age and weight

Spearman's rank correlation rho

```
data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]
S = 169075, p-value = 0.8857
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.01455125
```

Correlation between age and height

Spearman's rank correlation rho

```
data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]
S = 165305, p-value = 0.9365
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.008072461
```

Correlation between age and alcohol

Spearman's rank correlation rho

```
data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]
S = 184563, p-value = 0.2871
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.107488
```

Correlation between weight and height

Spearman's rank correlation rho

```
data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]
S = 11602, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.9303835
```

Correlation between weight and alcohol

Spearman's rank correlation rho

data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]

S = 152370, p-value = 0.3966

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.08568687

Correlation between height and alcohol

Spearman's rank correlation rho

data: numerical_data[[pair[1]]] and numerical_data[[pair[2]]]

S = 145752, p-value = 0.2138

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.1253977

Result – Correlation analysis Spearman's correlation analysis revealed a very strong positive association between weight and height ($\rho = 0.930$, p less than 0.001). This indicates that taller individuals tended to have higher body weight in this sample. No statistically significant correlations were observed between the other pairs of numerical variables (all p -values greater than 0.05). Overall, weight and height showed the strongest relationship among the variables examined.